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9.2.6 Jordan Canonical Form

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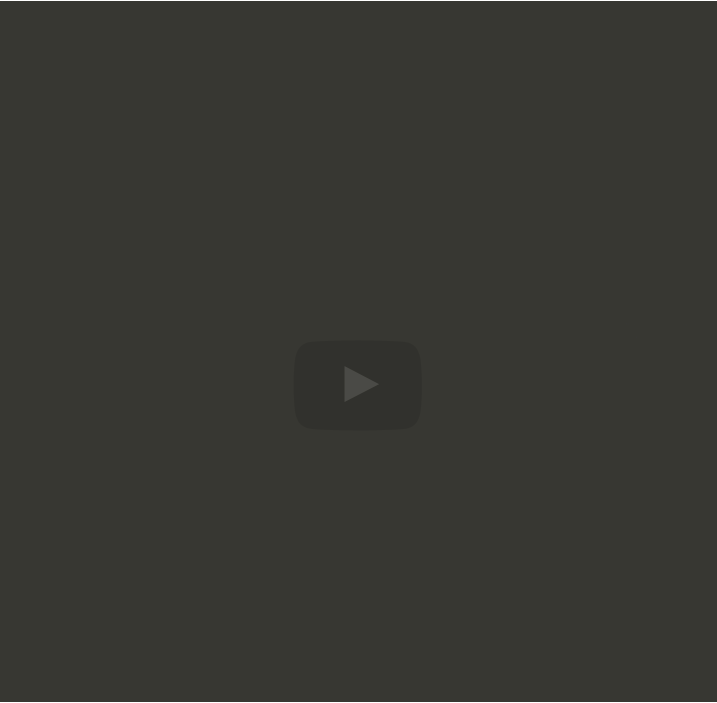
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Video



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diagonalizable is Jordan Block.

Okay, so what is this? This is a matrix that is k by k . On its diagonal you have μ , which obviously is it's only eigenvalue. And then on its super-diagonal you have all ones. And the rest of the matrix is all zeros. And I'm going to let you do the homework in which you compute what the eigenvalues and eigenvectors of this matrix are. And then we'll reconvene in the next video.

Video

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Reading assignment

1/1 point (graded)

Read Unit 9.2.6 of the notes. [\[LINK\]](#)

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Homework 9.2.5.1

1/1 point (graded)

Compute the eigenvalues of $k \times k$ matrix

$$J_k(\mu) = \begin{pmatrix} \mu & 1 & 0 & \cdots & 0 & 0 \\ 0 & \mu & 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \ddots & \mu & 1 \\ 0 & 0 & 0 & \cdots & 0 & \mu \end{pmatrix}$$

where $k > 1$. For each eigenvalue compute a basis for the subspace of its eigenvectors (including the zero vector to make it a subspace).

☒ done



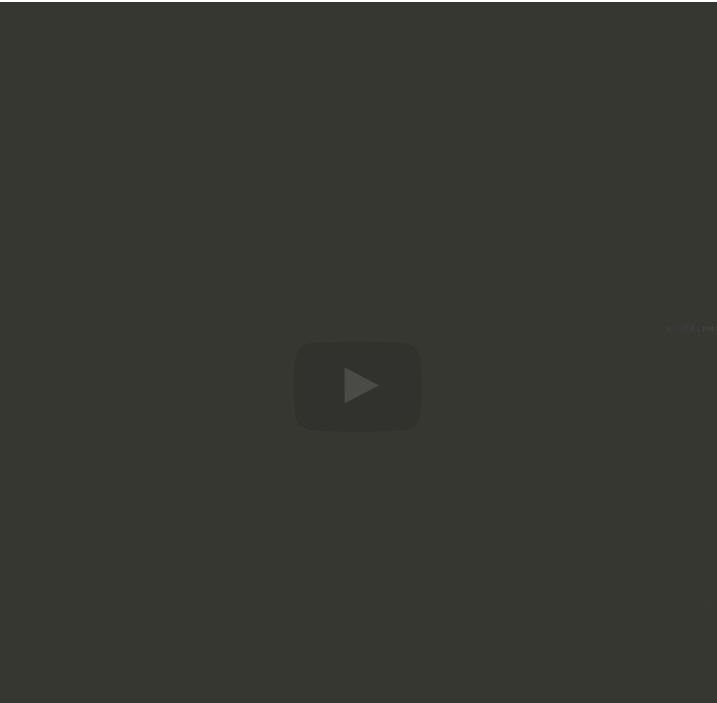
Solution:

For a complete solution see the [notes](#).

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Answers are displayed within the problem

Video



Okay. So how do we find the eigenvalues? Well I kind of gave it away already in the last video. It's just the mu on the diagonal. This particular matrix has only mu as its eigenvalue and that's an eigenvalue of multiplicity k. Now how do we find the eigenvectors

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1/1 point (graded)

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☒ done



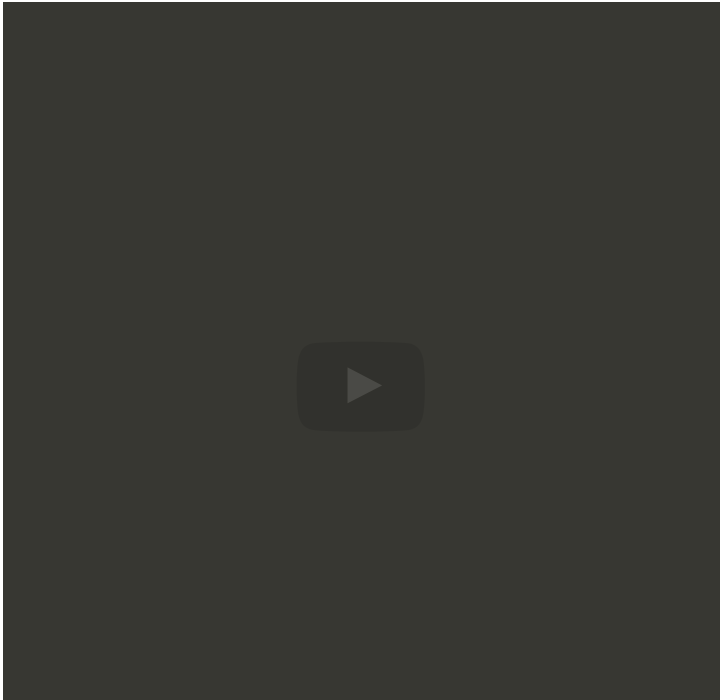
Solution:

For a complete solution see the [notes](#).

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... to that eigenvalue.
The total number of times that an eigenvalue appears along the diagonal of this matrix is then the algebraic multiplicity of the eigenvalue.
It's the multiplicity that that eigenvalue is a root of the determinant. The geometric multiplicity of an eigenvalue equals the number of Jordan blocks that have that eigenvalue as its eigenvalue.
In other words, it's equal to the number of linearly independent eigenvectors...

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